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(54) **DOWNLIGHT RETROFIT BRACKET**

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F21S 8/06 (2006.01)
F21V 23/00 (2015.01)

(52) **U.S. Cl.**
CPC ***F21V 21/047*** (2013.01); ***F21S 8/06*** (2013.01); ***F21V 21/04*** (2013.01); ***F21V 23/001*** (2013.01)

(58) **Field of Classification Search**
CPC F21V 21/04; F21V 21/047; F21V 23/001; F21S 8/06
USPC 362/147, 365, 366
See application file for complete search history.

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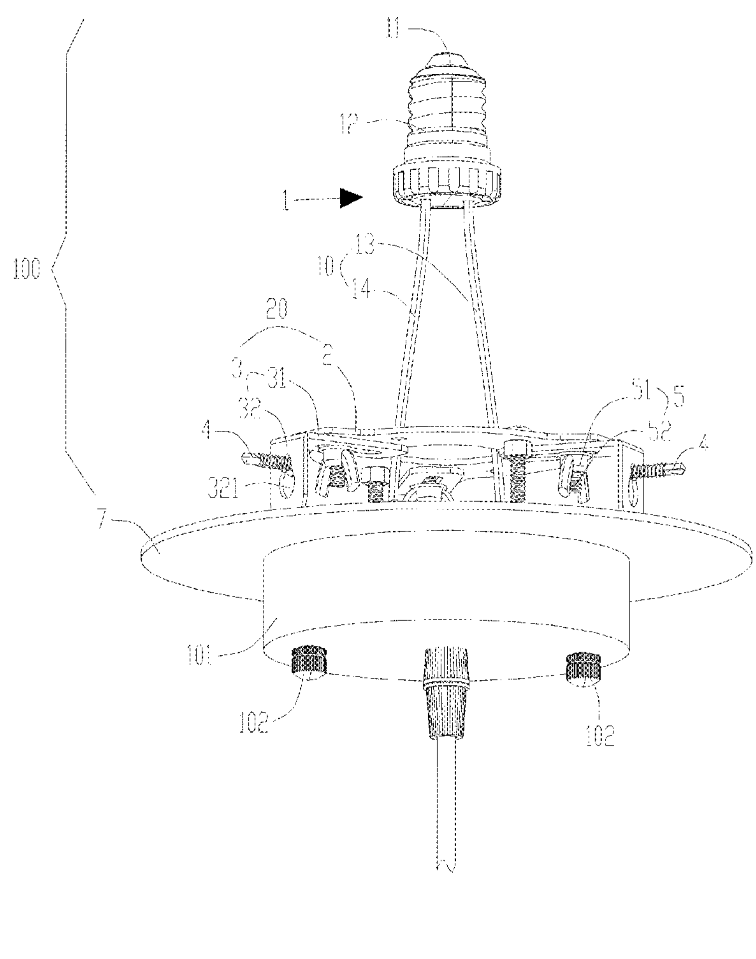
(57) **ABSTRACT**

The downlight retrofit bracket includes an adaptation base, a fixed bracket, a mounting plate, a height adjustment structure, and a decorative component. The adaptation base is configured to adapt to a lamp holder on a downlight, and a power cable electrically connected to the lamp holder on the downlight is arranged on the adaptation base, and without rewiring, so that a workload of retrofitting a light is reduced. The fixed bracket is arranged inside the downlight and is mechanically fixed to a ceiling. The mounting plate is configured to mechanically fix a suspended electrical appliance. The mounting plate is fixed to the fixed bracket through the height adjustment structure. The height adjustment structure is configured to be used to adjust an up-down relative position between the fixed bracket and the mounting plate. This improves the adaptability of the downlight retrofit bracket.

20 Claims, 7 Drawing Sheets



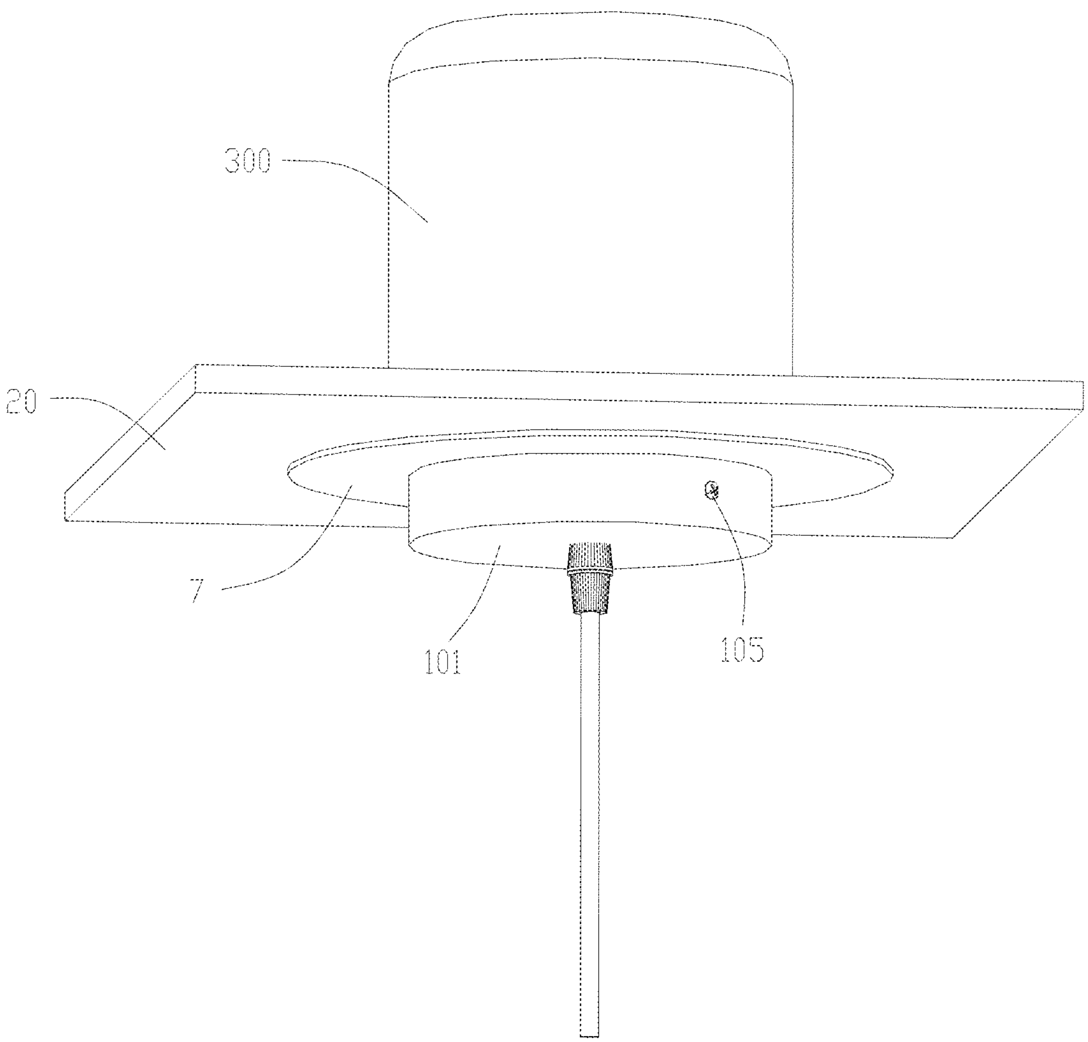


FIG. 1

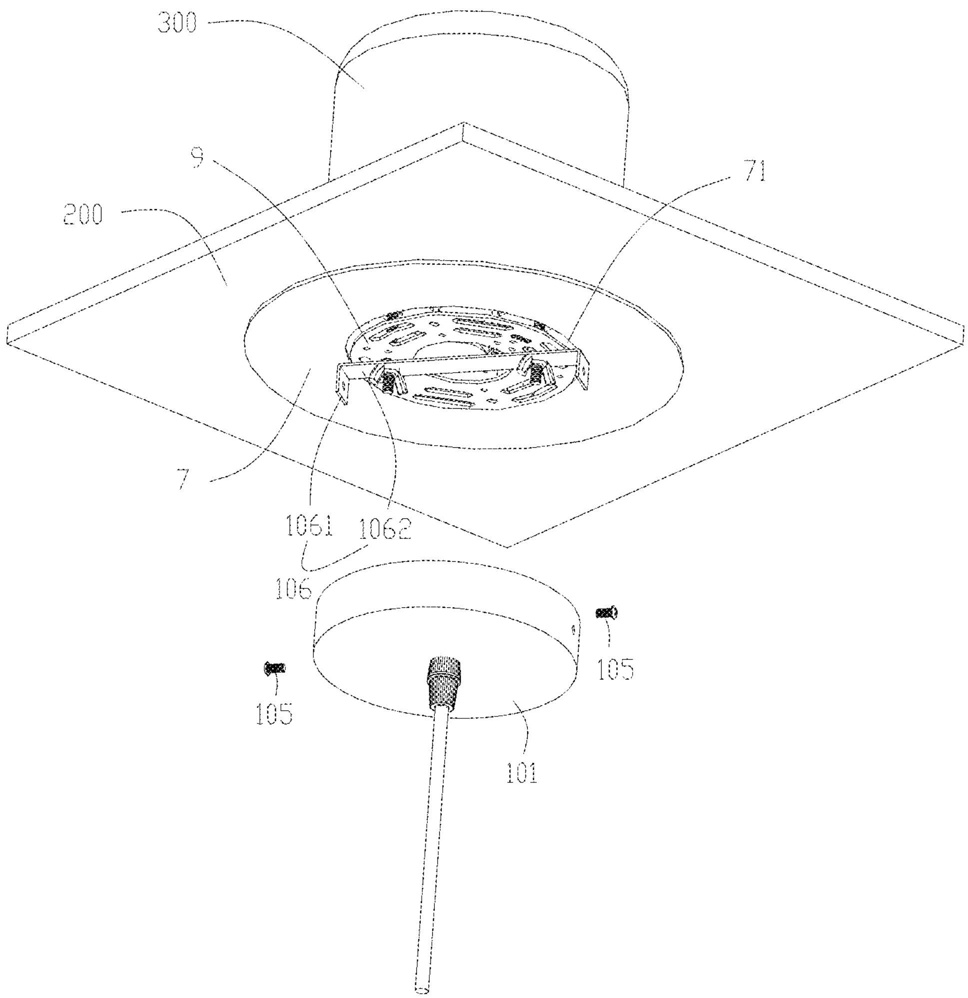


FIG. 2

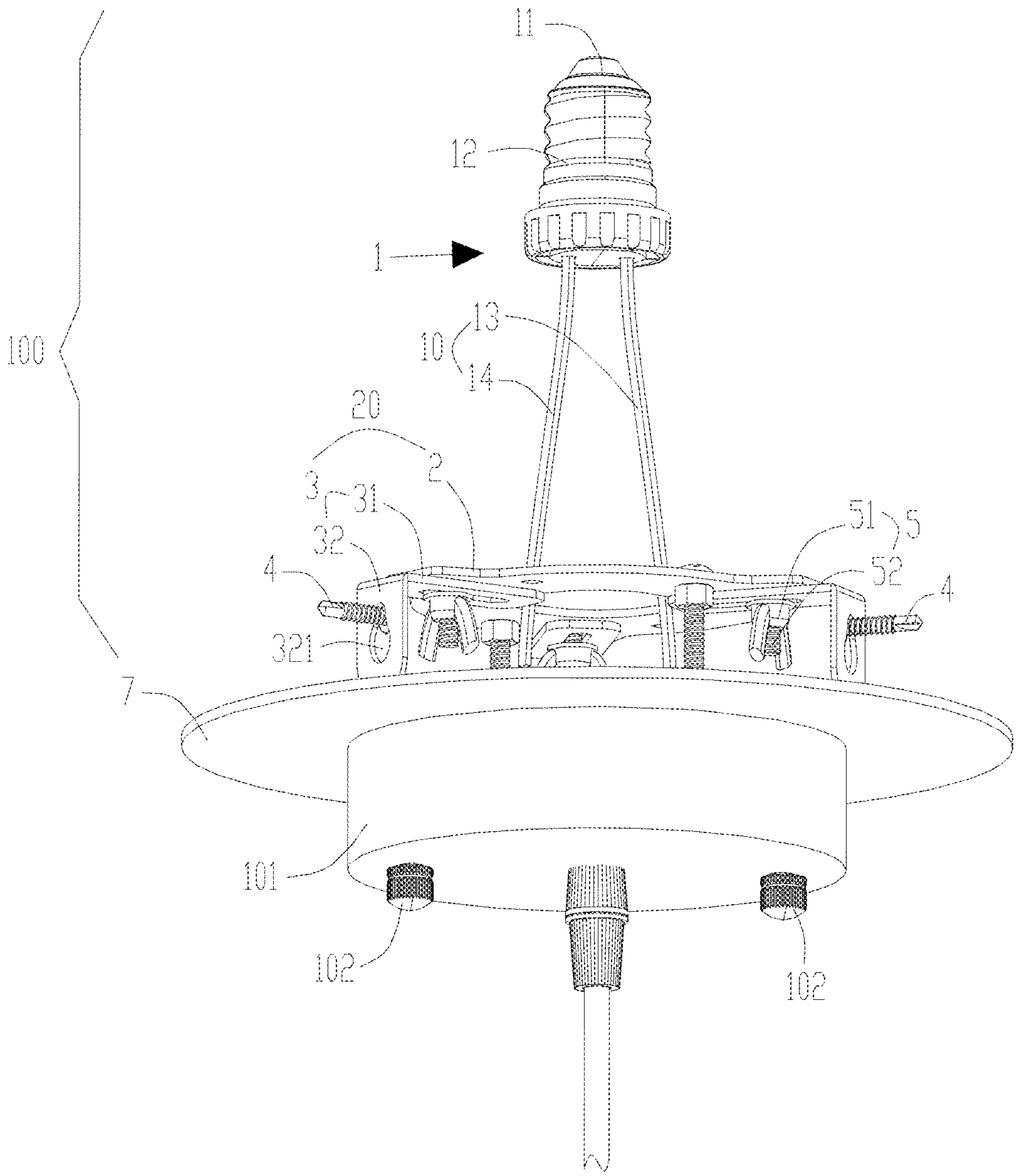
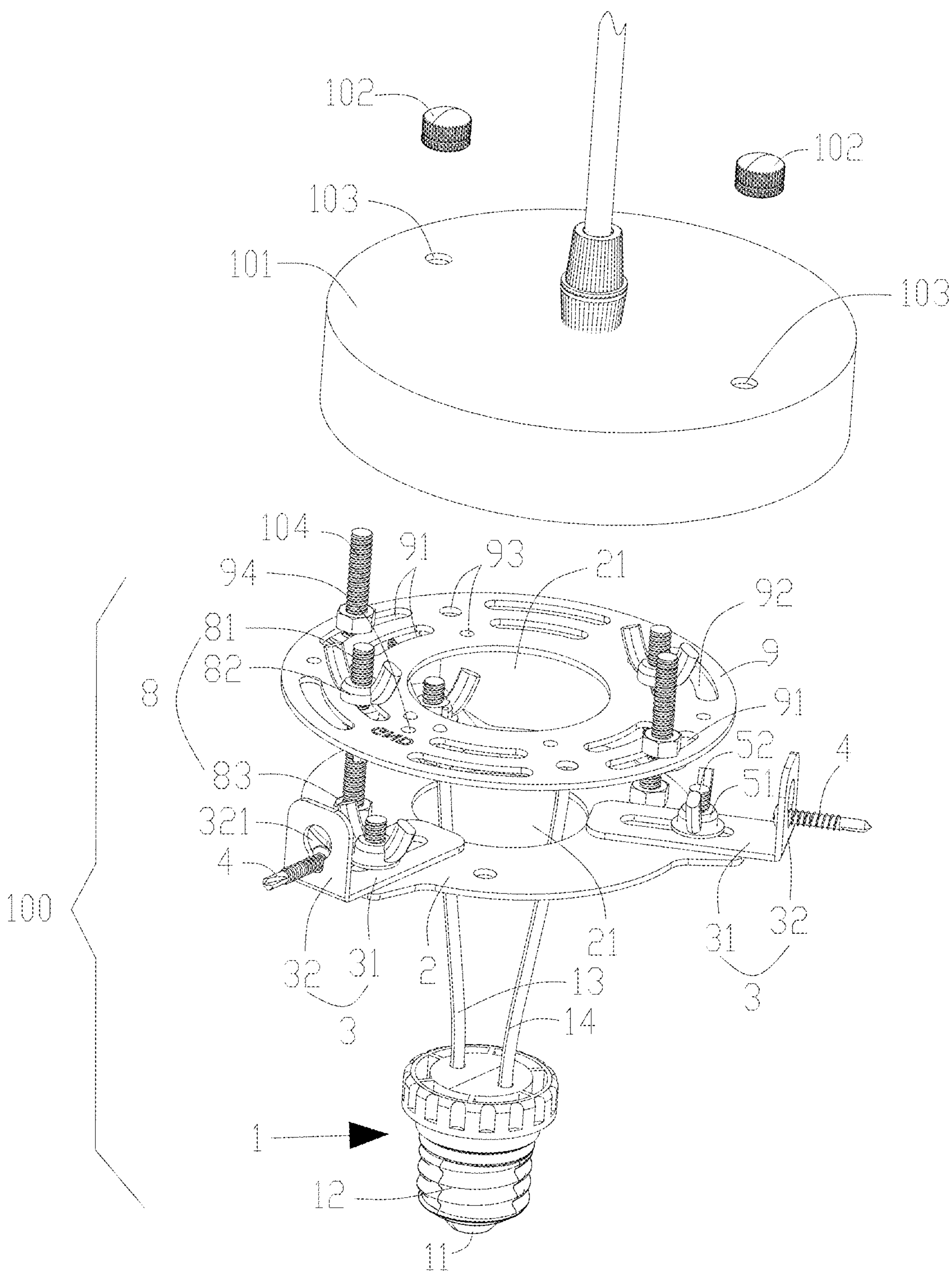


FIG. 3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102
103
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52
4
83
51
321
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100
2
21
31
32
32
31
31
3
13
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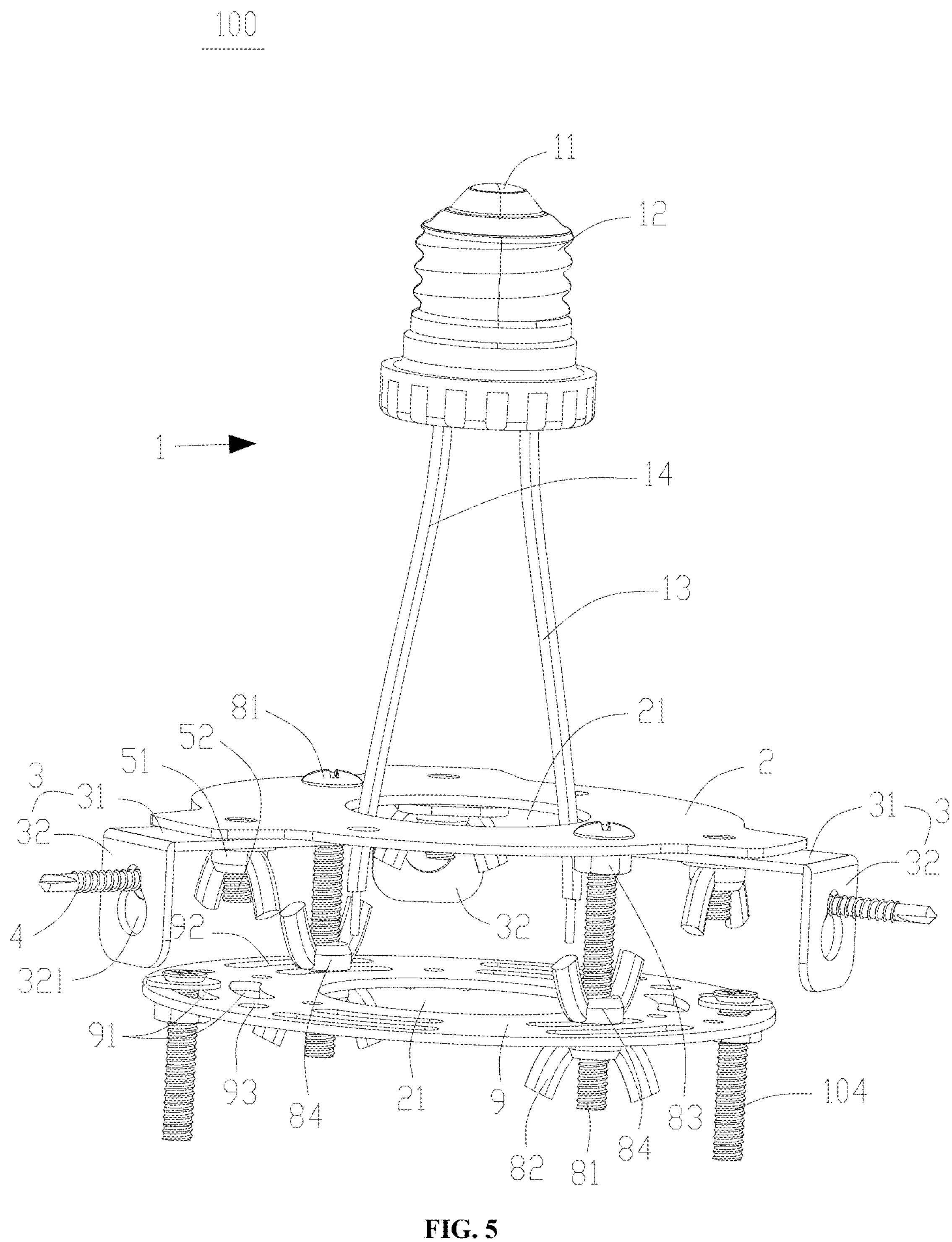
14
1
12
11

FIG. 4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11
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104
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FIG. 5

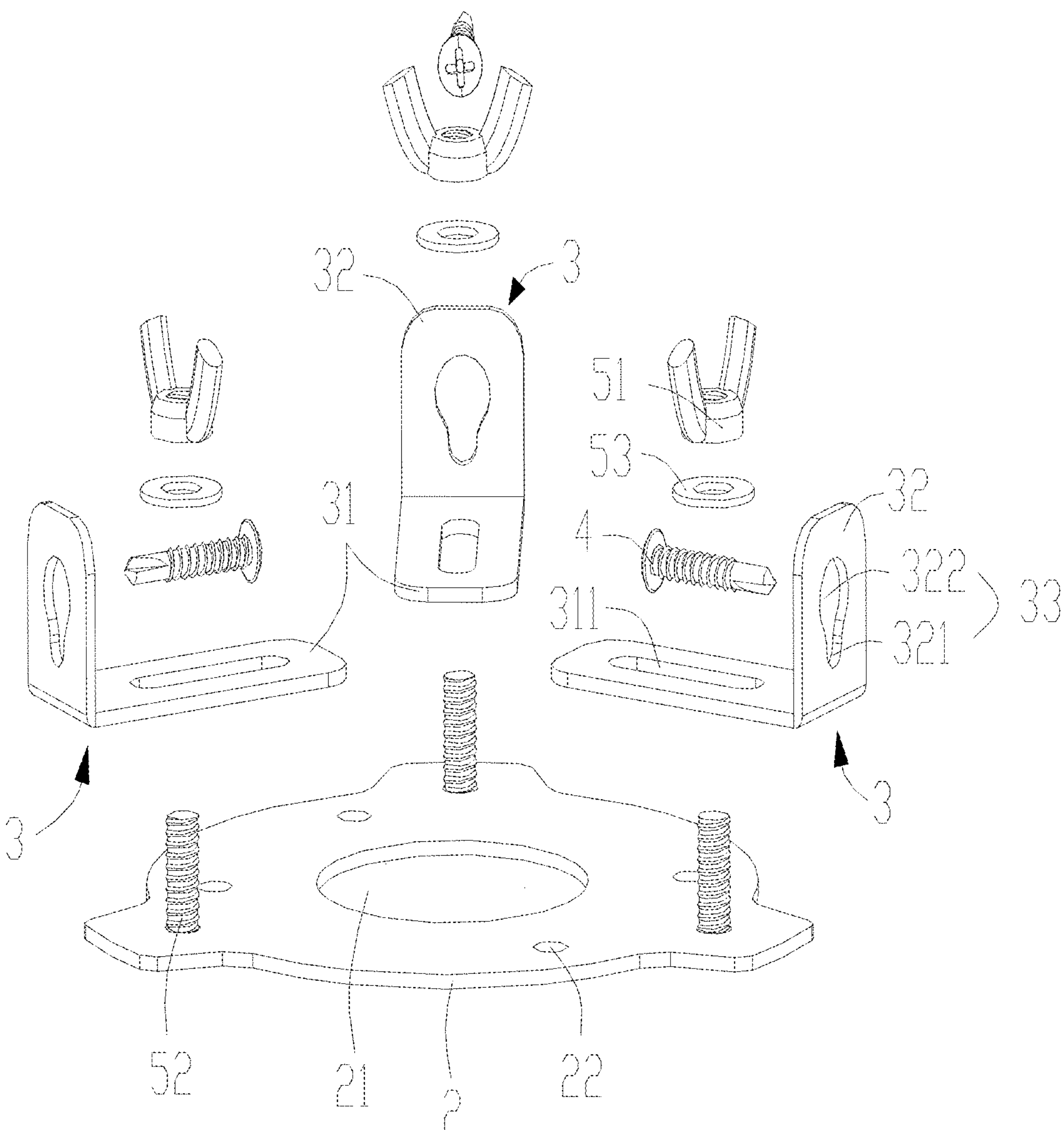

FIG. 6

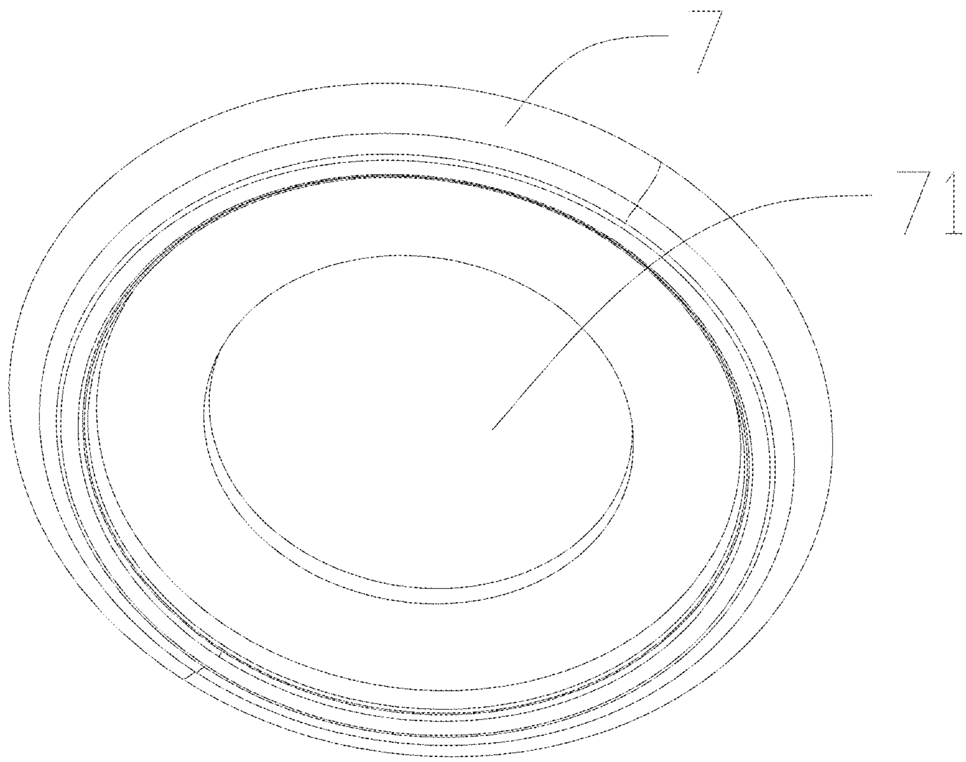

FIG. 7

DOWNLIGHT RETROFIT BRACKET

TECHNICAL FIELD

The present invention relates to the technical field of lamp brackets, and in particular, to a downlight retrofit bracket.

BACKGROUND

Since many residential buildings are constructed early, embedded spotlights (downlights) are commonly used for indoor lighting. The embedded spotlight includes a lamp tube fixed on a ceiling and a lamp holder arranged on the lamp tube. The lamp holder is used by a user to mount a bulb to achieve a lighting effect. However, the embedded spotlight has a limited illumination range, and the illumination brightness of the traditional bulb is limited. As a result, a lighting effect is poor, and a modern indoor lighting requirement cannot be met. Furthermore, with an outdated appearance, the embedded spotlight can be hardly integrated into a modern interior decoration style. In addition, when people replace or upgrade the embedded spotlight to a pendant light, people need to modify a ceiling and rearrange wires, so that a user needs to carry out large-scale indoor decoration, which is time-consuming, labor-intensive, and expensive.

SUMMARY

The present invention mainly aims to provide a downlight retrofit bracket, to solve the problem that replacing an embedded spotlight with a pendant light by a user is time-consuming, labor-intensive, and expensive.

In order to solve the technical problem, the technical scheme provided by the present invention is as follows.

A downlight retrofit bracket, including includes:

- an adaptation base for use with a lamp holder on a downlight, the adaptation base is provided with a power cable electrically connected to the lamp holder on the downlight;
- a fixed bracket arranged inside the downlight, the fixed bracket is mechanically fixed to a ceiling;
- a mounting plate for mechanically fixing a suspended electrical appliance, the mounting plate and the fixed bracket are provided with a first opening for threading out the power cable;
- a height adjustment structure, the mounting plate is fixed on the fixed bracket through the height adjustment structure, and the height adjustment structure is configured to be used to adjust an up-down relative position between the fixed bracket and the mounting plate; and
- a decorative component attached to a surface of the ceiling and configured for decoration.

Furthermore, the fixed bracket includes: a fixed plate spaced apart from the mounting plate in an up-down manner, and a plurality of connecting pieces arranged on the fixed plate; the fixed plate is provided with the first opening; the plurality of connecting pieces are distributed around a center on the fixed plate; and the connecting pieces are fixed to the ceiling through first fasteners.

Furthermore, each connecting piece includes a first connecting plate and a second connecting plate formed by bending the first connecting plate downward; the second connecting plate is fixed to the ceiling through the first fastener;

the fixed bracket further includes a second fastener; the first connecting plate is connected to the fixed plate through the second fastener; and the second fastener is at different positions on the first connecting plate or the fixed plate to adjust a relative position between the first connecting plate and the fixed plate.

Furthermore, a surface of the first connecting plate presses against a surface of the fixed plate, and the second connecting plate is perpendicular to the first connecting plate.

Furthermore, the first connecting plate is provided with an elongated strip-shaped hole; the second fastener includes a first screw and a first nut; the first screw is fixed on the fixed plate; the first screw passes through the strip-shaped hole and is locked by the first nut to fix the first connecting plate with the fixed plate; and a distance between two adjacent second connecting plates is defined by a relative position between the first screw and the strip-shaped hole.

Furthermore, the first screw is integrally formed with the fixed plate.

Furthermore, three connecting pieces are included, which are uniformly distributed around the center of the fixed plate.

Furthermore, the second connecting plate is provided with a first through hole, and the first fastener passes through the first through hole to fix the second connecting plate to the ceiling.

Furthermore, the second through hole includes a first hole body and a second hole body communicated to the first hole body; the first hole body and the second hole body are in a shape of 8; the first hole body is located below the second hole body; and a diameter of the first hole body is less than a diameter of the second hole body.

Furthermore, the height adjustment structure includes at least two second screws and second nuts adapted to be used with the second screws; the fixed plate is provided with a plurality of second through holes arranged in a surrounding manner; the mounting plate is provided with a plurality of third through holes arranged in a surrounding manner; the second screws pass through the second through holes and the third through holes and are locked by the second nuts to fix the mounting plate to the fixed plate; the second nuts are locked at different positions on the second screws to limit a distance between the mounting plate and the fixed plate; and the at least two second screws are distributed around an outer side of the first opening.

Furthermore, the third through holes are elongated.

Furthermore, the mounting plate is a circular panel; the first opening is located in a center position of the circular panel; the third through holes are arc-shaped; a circle center of each third through hole is consistent with a circle center of the circular panel; and the third through holes are located next to the first opening.

Furthermore, the plurality of third through holes are divided into two groups and distributed around a circumferential side of the first opening; and distances between the third through holes in the two groups and the first opening are not equal.

Furthermore, the mounting plate is further provided with a plurality of fourth through holes distributed around the outer side of the first opening; and the fourth through holes are staggered from the third through holes in a direction away from the first opening.

Furthermore, the height adjustment structure further includes third nuts; and the second screws pass through the second through holes and are locked by the third nuts to define a relative position between the second screws and the fixed plate.

Furthermore, the height adjustment structure further includes fourth nuts; and the fourth nuts are locked onto the second screws and cooperate with the second nuts to fix the mounting plate.

Furthermore, the mounting plate is further provided with a plurality of pendant light through holes for mounting a pendant light.

Furthermore, the mounting plate is provided with a ground wire hole for fixing a ground wire with a screw; and the mounting plate, the fixed bracket, and the height adjustment structure all have conductive properties.

Furthermore, the decorative component is provided with a second opening for exposing the mounting plate, or the decorative component is integrally injection-molded with the mounting plate.

Furthermore, a width of the second opening is less than a width of a pendant light ceiling plate, and the width of the second opening is greater than a width of the mounting plate.

Compared with the prior art, the present invention has the beneficial effects below. In this invention, the adaptation base is adaptively used with the lamp holder on the downlight, thereby fixing the adaptation base to the lamp holder of the downlight and electrically connecting the adaptation base to the lamp holder of the downlight. In conjunction with the power cable, the power cable leads out power of the lamp holder on the downlight through the adaptation base. After passing through the first openings, the power cable can supply power to a suspended electrical appliance, to cause the suspended electrical appliance to work normally, without removal of the original downlight, circuit change, or rewiring, so that a workload of retrofitting a light is reduced. Furthermore, by configuring the fixed bracket inside the downlight and mechanically fixing the fixed bracket to the ceiling, the height adjustment structure is used to fixedly connect the fixed bracket to the mounting plate and is used to adjust the up-down relative position between the fixed bracket and the mounting plate, so that the downlight retrofit bracket of this embodiment can adapt to pendant light ceiling plates with different heights and downlights with different sizes. This improves the adaptability of the downlight retrofit bracket. In addition, by configuring the decorative component for decoration, the downlight retrofit bracket can adapt to different styles of pendant lights when fixed to the ceiling for use, thus achieving better coordination between the mounted pendant light and a style of an environment.

BRIEF DESCRIPTION OF THE DRAWINGS

In order to explain the technical solutions of the embodiments of the present invention more clearly, the following will briefly introduce the accompanying drawings used in the embodiments. Apparently, the drawings in the following description are only some embodiments of the present invention. Those of ordinary skill in the art can obtain other drawings based on these drawings without creative work.

FIG. **1** is a schematic diagram of fixing in conjunction with a downlight, a ceiling, and a pendant light ceiling plate according to the present invention, where the pendant light ceiling plate is mounted on a side surface;

FIG. **2** is a partially exploded view of fixing in conjunction with a downlight, a ceiling, and a pendant light ceiling plate according to the present invention, where the pendant light ceiling plate is mounted on a side surface;

FIG. **3** is a schematic diagram of fixing in conjunction with a pendant light ceiling plate according to the present invention, where the pendant light ceiling plate is mounted on a surface;

FIG. **4** is a partially exploded view of fixing in conjunction with a pendant light ceiling plate according to the present invention, where the pendant light ceiling plate is mounted on a surface;

FIG. **5** is a structural diagram of fixing in conjunction with third screws according to the present invention;

FIG. **6** is an exploded view of a mounting bracket and a first fastener according to the present invention; and

FIG. **7** is a schematic diagram of a decorative component according to another embodiment of the present invention.

DETAILED DESCRIPTION OF THE EMBODIMENTS

The technical solutions in the embodiments of the present invention will be clearly and completely described below in conjunction with the accompanying drawings in the embodiments of the present invention. Apparently, the described embodiments are only a part of the embodiments of the present invention, rather than all the embodiments. Based on the embodiments in the present invention, all other embodiments obtained by those ordinarily skilled in the art without doing creative work shall fall within the protection scope of the present invention.

Referring to FIG. **1** to FIG. **7**, an embodiment of the present invention provides a downlight retrofit bracket **100**.

The downlight retrofit bracket **100** includes an adaptation base **1**, a fixed bracket **20**, a mounting plate **9**, a height adjustment structure **8**, and a decorative component **7**. The adaptation base **1** is configured to adapt to a lamp holder on a downlight **300**, and a power cable **10** electrically connected to the lamp holder on the downlight **300** is arranged on the adaptation base **1**. The fixed bracket **20** is arranged inside the downlight **300** and is mechanically fixed to a ceiling **200**. The mounting plate **9** is configured to mechanically fix a suspended electrical appliance. The mounting plate **9** is fixed to the fixed bracket **20** through the height adjustment structure **8**. The height adjustment structure **8** is configured to be used to adjust an up-down relative position between the fixed bracket **20** and the mounting plate **9**. The mounting plate **9** and the fixed bracket **20** are provided with a first opening **21** for threading out the power cable **10**. The decorative component **7** is fixed on the ceiling **200**.

In this embodiment, the adaptation base **1** is adaptively used with the lamp holder on the downlight **300**, thereby fixing the adaptation base **1** to the lamp holder of the downlight **300** and electrically connecting the adaptation base **1** to the lamp holder of the downlight **300**. In conjunction with the power cable **10**, the power cable **10** leads out power of the lamp holder on the downlight **300** through the adaptation base **1**. After passing through the first openings **21**, the power cable **10** can supply power to a suspended electrical appliance, to cause the suspended electrical appliance to work normally, without removal of the original downlight, circuit change, or rewiring, so that a workload of retrofitting a light is reduced. Furthermore, by configuring the fixed bracket **20** inside the downlight and mechanically fixing the fixed bracket **20** to the ceiling **200**, the height adjustment structure **8** is used to fixedly connect the fixed bracket **20** to the mounting plate **9** and is used to adjust the up-down relative position between the fixed bracket **20** and the mounting plate **9**, so that the downlight retrofit bracket **100** of this embodiment can adapt to pendant light ceiling plates **101** with different heights and downlights **300** with different sizes. This improves the adaptability of the downlight retrofit bracket **100**. In addition, by configuring the decorative component **7** for decoration, the downlight retrofit bracket **100** can adapt to different styles of pendant lights when fixed to the ceiling **200** for use, thus achieving better coordination between the mounted pendant light and a style of an environment.

Specifically, the suspended electrical appliance includes a ceiling fan, a pendant light, a track light, a light exposed out of the ceiling **200**, and the like. When the lamp holder on the existing lamp tube is E**26**, the adaptation base **1** is a lamp base that is matched with the lamp holder E**26**. When the lamp holder on the existing downlight **300** is E**27**, the adaptation base **1** is a lamp base that is matched with the lamp holder E**27**. In the lamp holder E**26** or E**27** on the existing lamp tube, a bottom surface of an inner surface of the lamp holder is a negative electrode, and an inner side wall of the lamp holder is a positive electrode. That is, an outer side wall of the adaptation base **1** is a positive electrode **12**, and a top surface of an outer surface of the adaptation base **1** is a negative electrode **11**. The power cable **10** is divided into a positive wire **13** and a negative wire **14**. The positive wire **13** is electrically connected to the positive electrode **12** of the adaptation base **1**, and the negative wire **14** is electrically connected to the negative electrode **11** of the adaptation base **1**, to lead out the power of the downlight retrofit bracket **100** in this embodiment without rewiring.

In an embodiment, the decorative component **7** is provided with a second opening **71** for exposing the mounting plate **9**. Certainly, in other embodiments, the decorative component **7** can also be integrally injection-molded as with the mounting plate **9**.

In an embodiment, the fixed bracket **20** includes a fixed plate **2** spaced apart from the mounting plate **9** in an up-down manner, and a plurality of connecting pieces **3** arranged on the fixed plate **2**. The fixed plate **2** is provided with the first opening **21**. The plurality of connecting pieces **3** are distributed around a center on the fixed plate **2**. The connecting pieces **3** are fixed to the ceiling **200** through first fasteners **4**, to fix the mounting plate **9** to the ceiling **200**. Specifically, the center is a center of the first opening **21** or the fixed plate **2**.

In an embodiment, each connecting piece **3** includes a first connecting plate **31** and a second connecting plate **32** formed by bending the first connecting plate **31** downward. The second connecting plate **32** is fixed to the ceiling **200** through the first fastener **4**. The fixed bracket **20** further includes a second fastener **5**. The first connecting plate **31** is connected to the fixed plate **2** through the second fastener **5**. The second fastener **5** is at different positions on the first connecting plate **31** or the fixed plate **2** to adjust a relative position between the first connecting plate **31** and the fixed plate **2**, thus adjusting a width of the entire fixed bracket **20** to adapt to widths of different lamp tubes.

Specifically, a surface of the first connecting plate **31** presses against a surface of the fixed plate **2**, to improve the compactness of the structure of the fixed bracket **20**. The second connecting plate **32** is perpendicular to the first connecting plate **31**. Certainly, in other embodiments, the second connecting plate **32** can also be arc-shaped, that is, it is not perpendicular to the first connecting plate **31**, and can be fixed to the ceiling **200** through the first fastener **4**.

In an embodiment, the first connecting plate **31** is provided with an elongated strip-shaped hole **311**. The second fastener **5** includes a first screw **52** and a first nut **51**. The first screw **52** is fixed on the fixed plate **2**. The first screw **52** passes through the strip-shaped hole **311** and is locked by the first nut **51** to fix the first connecting plate **31** with the fixed plate **2**. A distance between two adjacent second connecting plates **32** is defined by a relative position between the first screw **52** and the strip-shaped hole **311**. By using different positions of the first screw **52** on the strip-shaped hole **311**, the relative position between the first connecting plate **31** and the fixed plate **2** can be adjusted, thereby adjusting an overall width of the fixed bracket **20** to adapt to a width of the lamp tube, so that the second connecting plate **32** abuts against a side wall of the lamp tube, to stably fix the second connecting plate **32** to the ceiling **200** through the fastener that passes through the side wall of the lamp tube.

In an embodiment, the first screw **52** is integrally formed with the fixed plate **2** to reduce a quantity of parts for the downlight retrofit bracket **100** in this embodiment, and it is convenient for a user to mount the fixed plate **2** and the connecting piece **3**. Certainly, in other embodiments, the first screw **52** is not integrally formed with the fixed plate **2**. That is, the first screw **52** is locked by the first nut **51** after passing through the fixed plate **2** and the strip-shaped hole **311**, which can also adjust the relative position between the fixed plate **2** and the first connecting plate **31**.

In an embodiment, a gasket **53** is further arranged between the first nut **51** and the first connecting plate **31**, and the gasket **53** sleeves the first screw **52**. A width of the gasket **53** is greater than a width of the strip-shaped hole **311**. Using the gasket **53** can further improve the stability and smoothness of connection between the fixed plate **2** and the first connecting plate **31**.

In an embodiment, three connecting pieces **3** are included, which are uniformly distributed around the center of the fixed plate **2**. Certainly, in other embodiments, four, five, or more connecting pieces **3** may also be included.

In an embodiment, the second connecting plate **32** is provided with a first through hole **33**, and the first fastener **4** passes through the first through hole **33** to fix the second connecting plate **32** to the ceiling **200**, to thread one end of the first fastener **4** through the second connecting plate **32**.

In an embodiment, the first through hole **33** includes a first hole body **321** and a second hole body **322** communicated to the first hole body **321**. The first hole body **321** and the second hole body **322** are in a shape of figure "8". The first hole body **321** is located below the second hole body **322**. A diameter of the first hole body **321** is less than a diameter of the second hole body **322**. Furthermore, when the first fastener **4** is a nail or a screw, a width of a head portion of the nail and a width of a head portion of the screw are both larger than the diameter of the first hole body **321**, but less than the diameter of the second hole **322**. That is, a user can first fix the screw and the nail to the ceiling **200**, and then align the second hole body **322** with the screw or the nail, thus hanging the fixed bracket **20** on the first fastener **4**.

Certainly, in other embodiments, the connecting piece **3** can also be integrally formed with the fixed plate **2**. Furthermore, a plurality of strip-shaped holes **311** may be included, which are fixedly distributed and spaced apart in a length direction of the first connecting plate **31**.

In an embodiment, the height adjustment structure **8** includes at least two second screws **81** and second nuts **82** adapted to be used with the second screws **81**. The fixed plate **2** is provided with a plurality of second through holes **22** arranged in a surrounding manner. The mounting plate **9** is provided with a plurality of third through holes **92** arranged in a surrounding manner. The second screws **81** pass through the second through holes **22** and the third through holes **92** and are locked by the second nuts **82** to fix the mounting plate **9** to the fixed plate **2**. The second nuts **82** are locked at different positions on the second screws **81** to limit a distance between the mounting plate **9** and the fixed plate **2**. The at least two second screws **81** are distributed around an outer side of the first opening **21**. The fixed plate **2** is fixed to the ceiling **200** through the connecting pieces **3**. That is, the position of the fixed plate **2** remains unchanged, and the second screws **81** are basically perpendicular to the fixed plate **2**. By adjusting relative positions of the second nuts **82** on the second screws **81**, the distance between the mounting plate **9** and the fixed plate **2** can be adjusted, so that a user can adjust the distance between the mounting plate **9** and the fixed plate **2** independently according to different pendant light ceiling plates **101** and different lamp tube depths, to adapt to different pendant light ceiling plates **101** and different lamp tube depths for use.

In an embodiment, the third through holes **92** are elongated, so that a user can sleeve the second screws **81** with the third through holes **92** in the mounting plate **9** and fix the mounting plate **9** with the second nuts **82**. The user does not need to accurately align the third through holes **92** with the second screws **81**. The elongated third through holes **92** allows for an offset in relative positions between the plurality of second screws **81**, making it convenient for mounting. Moreover, by using the elongated third through holes **92**, it is possible to ensure that the second nuts **82** on the plurality of second screws **81** are not located on the same plane, that is, heights of the second nuts **82** on the plurality of second screws **81** are not consistent, so that the mounting plate **9** is inclined. Namely, the mounting plate **9** and the fixed plate **2** are not parallel to each other, to adapt to the ceiling **200** with an inclination angle. This improves the practicality of the downlight retrofit bracket **100** in this embodiment.

In an embodiment, the mounting plate **9** is a circular panel. The first opening **21** is located in a center position of the circular panel. The third through holes **92** are arc-shaped. A circle center of each third through hole **92** is consistent with a circle center of the circular panel. The third through holes **92** are located next to the first opening **21**.

In an embodiment, the plurality of third through holes **92** are distributed in two rows, and the third through holes **92** in each row are distributed around the circle center of the circular panel. That is, the plurality of third through holes **92** are divided into two groups which are distributed around a circumferential side of the first opening **21**. Distances between the third through holes **92** in the two groups and the first opening **21** are not equal, so that the third through holes **92** at different positions can be selected to adapt to the pendant light ceiling plates **101** with different widths. This allows a user to fix the mounting plate **9** through cooperation between the third through holes **92** and the second screws **81**, without accurately aligning the third through holes **92** with the second screws **81**, making it convenient for mounting.

In an embodiment, 12 third through holes **92** are included, and a quantity of the third through holes **92** in each row is equal. That is, each row includes six third through holes **92**, so that a user can select at least two third through holes **92** to fix a pendent light ceiling plate together with screws and nuts.

Certainly, in other embodiments, when the third through holes **92** are dot-like, i.e. are not elongated, at least four third through holes **92** are included, and two third through holes **92** can also be reserved to fix a pendent light ceiling plate together with screws and nuts.

In an embodiment, the mounting plate **9** is further provided with a plurality of fourth through holes **93** distributed around the outer side of the first opening **21**. The fourth through holes **93** and the third through holes **92** are arranged in a staggered direction in a direction away from the first opening **21**, that is, a distance between two symmetrical fourth through holes **93** and a distance between two symmetrical third through holes **92** are different, to improve the adaptability of the downlight retrofit bracket **100** in this embodiment to more pendant light ceiling plates **101** with different widths.

At present, most of the existing pendent light ceiling plates **101** are generally fixed to the ceiling **200** by using two screws, and a distance between the two screws is generally 90 cm, that is, the distance between the two symmetrical third through holes **92** can be 90 cm. Certainly, in a pendant light ceiling plate **101** with another size, the distance between the two screws that fix the pendant light ceiling plate **101** may be greater than or less than 90 cm. That is, by configuring the plurality of arc-shaped third through holes **92**, dot-like fourth through holes, and the like, the mounting plate **9** can fix pendant light ceiling plates **101** with different sizes.

In an embodiment, the mounting plate **9** is provided with a ground wire hole **94** for fixing a ground wire with a screw. The mounting plate **9**, the fixed bracket **20**, and the height adjustment structure **8** all have conductive properties. Some of existing pendent lights have three power connection wires, namely a positive electrode, a negative electrode, and a ground wire. The positive electrode and the negative electrode are respectively connected to the positive wire **13** and the negative wire **14** of the adaptation base **1**. The ground wire can be fixed in the ground wire hole **94** through a screw, to connect the mounting plate **9**, the height adjustment structure **8**, and the fixed bracket **20** to a ground side wall of the downlight or the ceiling **200** to achieve a grounding effect. Certainly, in an embodiment where the ground wire hole **94** is not configured, the ground wire can be directly placed on a wall of the downlight, the mounting plate **9**, or the like.

In an embodiment, the height adjustment structure **8** further includes third nuts **83**. The second screws **81** pass through the second through holes **22** and are locked by the third nuts **83** to define a relative position between the second screws **81** and the fixed plate **2**. To prevent the second screws **81** from moving relative to the fixed plate **2** during the mounting of a pendant light, which may affect the balance of the mounting plate **9**.

In an embodiment, the height adjustment structure **8** further includes fourth nuts **84**. The fourth nuts **84** are locked onto the second screws **81** and cooperate with the second nuts **82** to fix the mounting plate **9**. A height of the mounting plate **9** is displayed using the fourth nuts **84** to prevent such a situation that when a user mounts a pendent light ceiling plate **101** by using the mounting plate **9**, the mounting plate **9** moves upward, which affects the mounting.

In an embodiment, the mounting plate **9** is further provided with a plurality of pendant light through holes **91** for mounting a pendant light. The pendant light through holes **91** can be third through holes **92** or other through holes, to fix the pendant light ceiling plate **101** on the mounting plate **9** together with screws and nuts.

In an embodiment, a width of the second opening **71** is less than a width of the pendant light ceiling plate **101**. When the pendant light ceiling plate **101** is mechanically fixed on the mounting plate **9**, the decorative component **7** can be compressed and fixed to the ceiling **200**.

In an embodiment, the width of the second opening **71** is greater than a width of the mounting plate **9** to fully expose the mounting plate **9**, so that a user can mechanically fix the pendant light ceiling plate **101** to the mounting plate **9** through the third through holes **92** or the pendant light through holes **91**.

In an embodiment, the decorative component **7** is panel-shaped. The decorative component **7** is covered by the pendant light ceiling plate **101** when the pendant light ceiling plate **101** is mechanically fixed to the mounting plate **9**. Different styles of pendent lights can be configured with different decorative components **7** to achieve better coordination between a style of a mounted pendent light and a style of an environment. Certainly, in other embodiments, the decorative component **7** may not be panel-shaped, and may be a part with a shape or a concave-convex appearance, as shown in FIG. **7**.

Based on the downlight retrofit bracket **100** in the above embodiment, referring to FIG. **3** and FIG. **4**, a surface of a pendant lamp ceiling plate **101** is mounted on the bracket of this embodiment. A method for mounting the surface of the pendant light ceiling plate **101** includes the following steps:

1. Fixing the adaptation base **1** onto the lamp holder.
2. Threading the fixed plate **2** through the strip-shaped holes **311** on the first connecting plates **31** through the first screw **52**, and locking and fixing the fixed plate **2** through the first nuts **51** to fix the plurality of connecting pieces **3** around the first opening **21**; and fixing the second connecting plates **32** to the ceiling **200** through the first fasteners **4** to fix the fixed bracket **20**.
3. Fix the mounting plate **9** to the fixed plate **2** through the height adjustment structure **8**.
4. Connecting the power cable **10** to a wire on a pendent light.
5. Threading one end of each third screw **104** through each third through hole **92** or each pendant light through holes **91**; then threading one end of the third screw **104** through each via hole **103** of the pendant light ceiling plate **101** in a thickness direction of the pendant light ceiling plate **101**, and locking the end through each fifth nut **102** to fix the pendant light ceiling plate **101** and compress the decorative part **7** to fix the pendant light, thus completing retrofit of the downlight **300** into the pendant light.

In the steps of the above mounting method, orders of steps **1**, **2**, and **3** can be interchanged, which does not affect the mounting of the pendant light ceiling plate **101**. The decorative component **7** can also be preliminarily fixed to the ceiling **200** by pasting. In this way, the decorative component **7** can be mounted to the ceiling **200** in any step before step **4**. When the decorative component **7** is not fixed to the ceiling **200** in advance, the decorative component **7** sleeves an outer side of the mounting plate **9** or the height adjustment structure **8** between step **3** and step **4**, and is further compressed and fixed when the pendant light ceiling plate **101** is fixed.

Furthermore, based on the downlight retrofit bracket **100** in the above embodiment, referring to FIG. **1** and FIG. **2**, a side surface of a pendant lamp ceiling plate **101** is mounted on the bracket of this embodiment. A method for mounting the side surface of the pendant light ceiling plate **101** includes the following steps:

1. Fixing the adaptation base **1** onto the lamp holder.
2. Threading the fixed plate **2** through the strip-shaped holes **311** on the first connecting plates **31** through the first screw **52**, and locking and fixing the fixed plate **2** through the first nuts **51** to fix the plurality of connecting pieces **3** around the first opening **21**; and fixing the second connecting plates **32** to the ceiling **200** through the first fasteners **4** to fix the fixed bracket **20**.
3. Respectively threading the plurality of second screws **81** through the corresponding second through holes **22**, the corresponding pendant light through holes **91**, and a side surface mounting panel **1062** on a side surface mounting bracket **106**, and locking the second screws **81** through the second nuts **82**.
4. Connecting the power cable **10** to a wire on a pendent light.
5. Sleeving the side surface mounting bracket **106** with the pendant light ceiling plate **101**; placing one end of each of a plurality of fourth screws **105** from an outer side of the pendant light ceiling plate **101** to an inner side of the pendant light ceiling plate **101**; and locking the end to a side edge connecting plate **1061** on the side surface mounting bracket **106**, to fix the pendant light ceiling plate **101**, thus completing the fixing of the pendant light and completing retrofit of the downlight **300** into the pendant light.

In the steps of the above mounting method, orders of steps **1**, **2**, and **3** can be interchanged, which does not affect the mounting of the pendant light ceiling plate **101**. The decorative component **7** can also be preliminarily fixed to the ceiling **200** by pasting. In this way, the decorative component **7** can be mounted to the ceiling **200** in any step before step **4**. When the decorative component **7** is not fixed to the ceiling **200** in advance, the decorative component **7** sleeves an outer side of the mounting plate **9** or the height adjustment structure **8** between step **3** and step **4**, and is further compressed and fixed when the pendant light ceiling plate **101** is fixed.

Certainly, in an embodiment where the mounting plate **9** is integrated with the decorative component **7**, the mounting plate **9** can be directly fixed to the fixed plate **2** through the height adjustment structure **8**, which can eliminate the mounting of the decorative component **7**.

It should be noted that all directional indications (such as up, down, left, right, front, back . . .) in the embodiments of the present invention are only used to explain a relative positional relationship between components, motion situations, etc. at a certain specific attitude (as shown in the figures). If the specific attitude changes, the directional indication also correspondingly changes.

In addition, the descriptions of "first", "second", etc. in the present invention are only used for descriptive purposes, and cannot be understood as indicating or implying its relative importance or implicitly indicating the number of technical features indicated. Therefore, features defined by "first" and "second" can explicitly instruct or impliedly include at least one feature. In addition, "and/or" in the entire text includes three solutions. A and/or B is taken as an example, including technical solution A, technical solution B, and technical solutions that both A and B satisfy. In addition, the technical solutions between the various embodiments can be combined with each other, but it needs be based on what can be achieved by those of ordinary skill in the art. When the combination of the technical solutions is contradictory or cannot be achieved, it should be considered that such a combination of the technical solutions does not exist, and is not within the scope of protection claimed by the present invention.

The above descriptions are only preferred embodiments of the present invention, and are not intended to limit the patent scope of the present invention. Any equivalent structural transformation made by using the content of the specification and the drawings of the present invention under the invention idea of the present invention, directly or indirectly applied to other related technical fields, shall all be included in the scope of patent protection of the present invention.

What is claimed is:

1. A downlight retrofit bracket, comprising:
an adaptation base for use with a lamp holder on a downlight, wherein the adaptation base is provided with a power cable electrically connected to the lamp holder on the downlight;
a fixed bracket arranged inside the downlight, wherein the fixed bracket is mechanically fixed to a ceiling;
a mounting plate for mechanically fixing a suspended electrical appliance, wherein the mounting plate and the fixed bracket are provided with a first opening for threading out the power cable;
a height adjustment structure, wherein the mounting plate is fixed on the fixed bracket through the height adjustment structure, and the height adjustment structure is configured to be used to adjust an up-down relative position between the fixed bracket and the mounting plate; and
a decorative component configured to be attached to a surface of the ceiling for decoration.

2. The downlight retrofit bracket according to claim **1**, wherein the fixed bracket comprises: a fixed plate spaced apart from the mounting plate in an up-down manner, and a plurality of connecting pieces arranged on the fixed plate; the fixed plate is provided with the first opening; the plurality of connecting pieces are distributed around a center on the fixed plate; and the connecting pieces are fixed to the ceiling through first fasteners.

3. The downlight retrofit bracket according to claim **2**, wherein each connecting piece comprises a first connecting plate and a second connecting plate formed by bending the first connecting plate; the second connecting plate is fixed to the ceiling through the first fastener;
the fixed bracket further comprises a second fastener; the first connecting plate is connected to the fixed plate through the second fastener; and the second fastener is at different positions on the first connecting plate or the fixed plate to adjust a relative position between the first connecting plate and the fixed plate.

4. The downlight retrofit bracket according to claim **3**, wherein a surface of the first connecting plate presses against a surface of the fixed plate, and the second connecting plate is perpendicular to the first connecting plate.

5. The downlight retrofit bracket according to claim **3**, wherein the first connecting plate is provided with an elongated strip-shaped hole; the second fastener comprises a first screw and a first nut; the first screw is fixed on the fixed plate; the first screw passes through the strip-shaped hole and is locked by the first nut to fix the first connecting plate with the fixed plate; and a distance between two adjacent second connecting plates is defined by a relative position between the first screw and the strip-shaped hole.

6. The downlight retrofit bracket according to claim **5**, wherein the first screw is integrally formed with the fixed plate.

7. The downlight retrofit bracket according to claim **5**, wherein the plurality of connecting pieces comprises three connecting pieces, the three connecting pieces are uniformly distributed around the center of the fixed plate.

8. The downlight retrofit bracket according to claim **5**, wherein the second connection plate is provided with a first through-hole, and a second through-hole and the first fastener passes through the first through hole to fix the second connecting plate to the ceiling.

9. The downlight retrofit bracket according to claim **8**, wherein the first through hole comprises a first hole body and a second hole body communicated to the first hole body; and a diameter of the first hole body is less than a diameter of the second hole body.

10. The downlight retrofit bracket according to claim **2**, wherein the height adjustment structure comprises at least two second screws and second nuts adapted to be used with the second screws; the fixed plate is provided with a plurality of second through holes arranged in a surrounding manner; the mounting plate is provided with a plurality of third through holes arranged in a surrounding manner; the second screws pass through the second through holes and the third through holes and are locked by the second nuts to fix the mounting plate to the fixed plate; the second nuts are locked at different positions on the second screws to limit a distance between the mounting plate and the fixed plate; and the at least two second screws are distributed around an outer side of the first opening.

11. The downlight retrofit bracket according to claim **10**, wherein the third through holes are elongated.

12. The downlight retrofit bracket according to claim **10**, wherein the mounting plate is a circular panel; the first opening is located in a center position of the circular panel; the third through holes are arc-shaped; a circle center of each third through hole is consistent with a circle center of the circular panel; and the third through holes are located next to the first opening.

13. The downlight retrofit bracket according to claim **10**, wherein the plurality of third through holes are divided into two groups and distributed around a circumferential side of the first opening; and distances between the third through holes in the two groups and the first opening are not equal.

14. The downlight retrofit bracket according to claim **13**, wherein the mounting plate is further provided with a plurality of fourth through holes distributed around the outer side of the first opening; and the fourth through holes are staggered from the third through holes in a direction away from the first opening.

15. The downlight retrofit bracket according to claim **10**, wherein the height adjustment structure further comprises third nuts; and the second screws pass through the second through holes and are locked by the third nuts to define a relative position between the second screws and the fixed plate.

16. The downlight retrofit bracket according to claim **15**, wherein the height adjustment structure further comprises fourth nuts; and the fourth nuts are locked onto the second screws and cooperate with the second nuts to fix the mounting plate.

17. The downlight retrofit bracket according to claim **10**, wherein the mounting plate is further provided with a plurality of pendant light through holes for mounting a pendant light.

18. The downlight retrofit bracket according to claim **1**, wherein the mounting plate is provided with a ground wire hole for fixing a ground wire with a screw; and the mounting plate, the fixed bracket, and the height adjustment structure all have conductive properties.

19. The downlight retrofit bracket according to claim **1**, wherein the decorative component is provided with a second opening for exposing the mounting plate, or the decorative component is integrally injection-molded with the mounting plate.

20. The downlight retrofit bracket according to claim **19**, wherein a width of the second opening is less than a width of a pendant light ceiling plate, and the width of the second opening is greater than a width of the mounting plate.

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